

Clustering

Lecture 3: Hierarchical Methods

Jing Gao
SUNY Buffalo

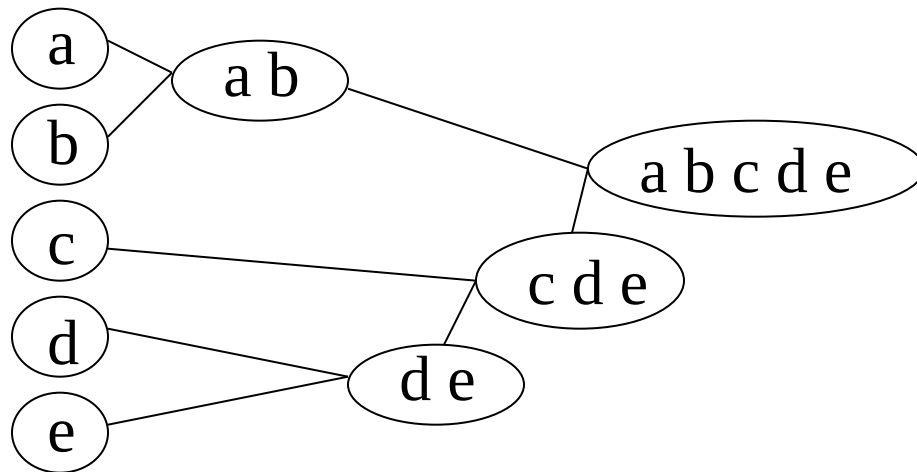
Up to Slide #25

Outline

- **Basics**
 - Motivation, definition, evaluation
- **Methods**
 - Partitional
 - Hierarchical
 - Density-based
 - Mixture model
 - Spectral methods
- **Advanced topics**
 - Clustering ensemble
 - Clustering in MapReduce
 - Semi-supervised clustering, subspace clustering, co-clustering, etc.

Hierarchical Clustering

- **Agglomerative approach**



Initialization:

Each object is a cluster

Iteration:

Merge two clusters which are most similar to each other;

Until all objects are merged into a single cluster

Step 0

Step 1

Step 2

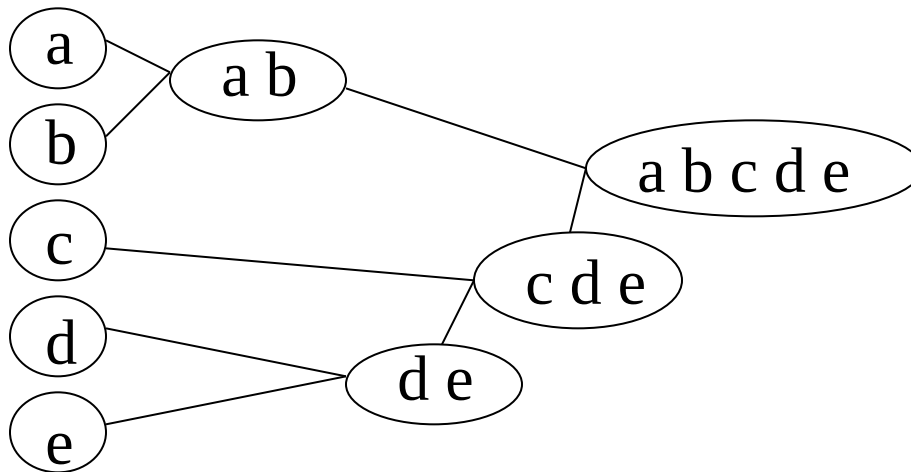
Step 3

Step 4

bottom-up

Hierarchical Clustering

- **Divisive Approaches**



Initialization:

All objects stay in one cluster

Iteration:

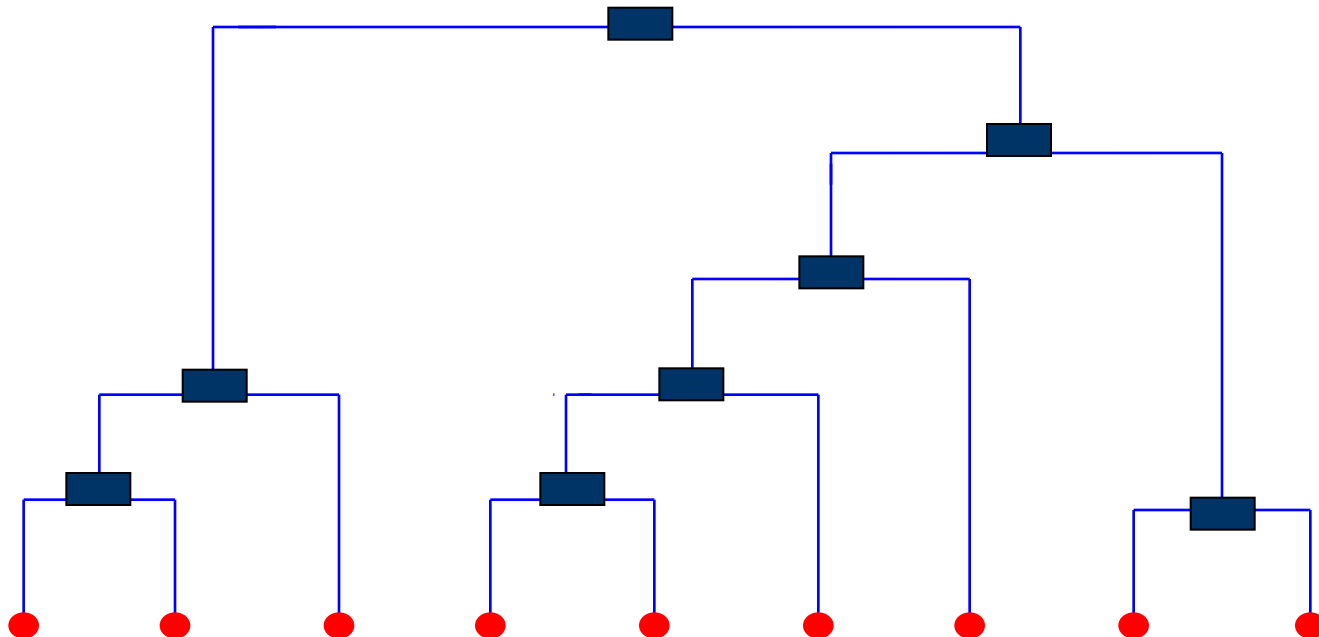
Select a cluster and split it into
two sub clusters

Until each leaf cluster contains
only one object

← Step 4 Step 3 Step 2 Step 1 **Step 0** Top-down

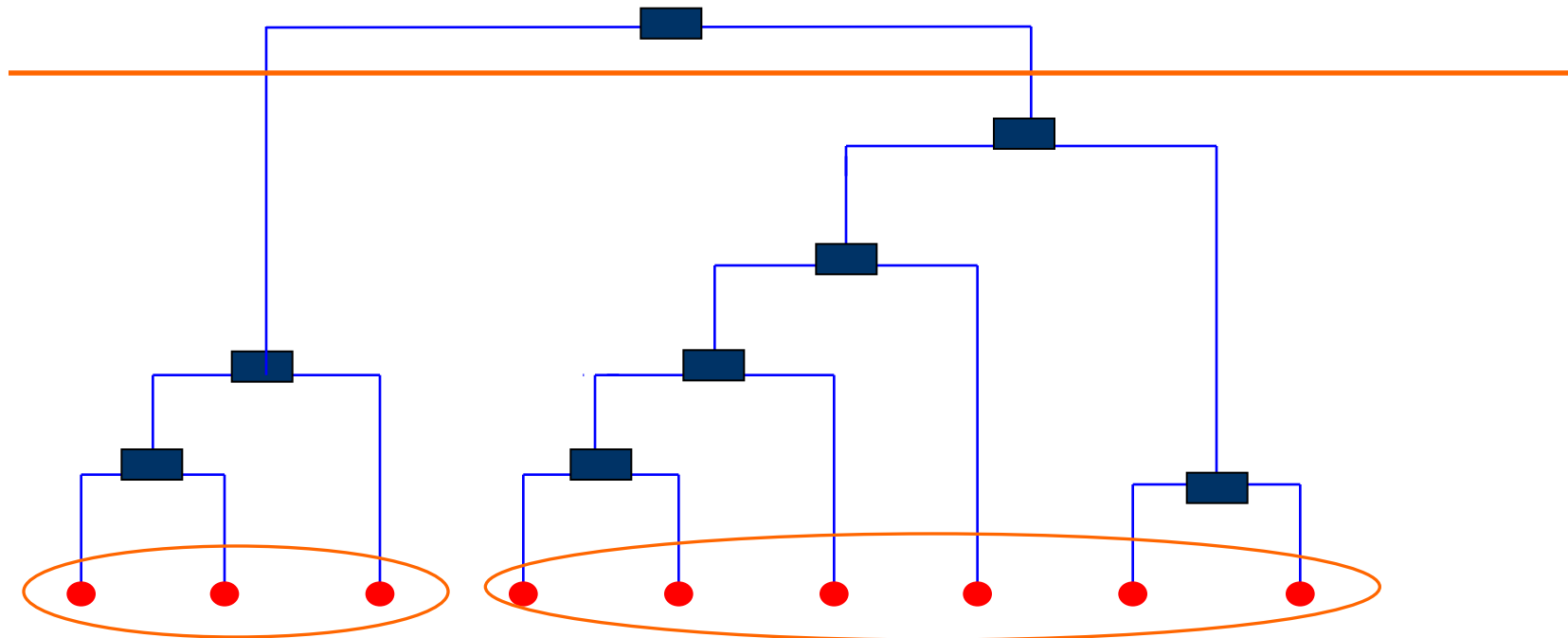
Dendrogram

- A tree that shows how clusters are merged/split hierarchically
- Each node on the tree is a cluster; each leaf node is a singleton cluster



Dendrogram

- A clustering of the data objects is obtained by cutting the *dendrogram* at the desired level, then each connected component forms a cluster

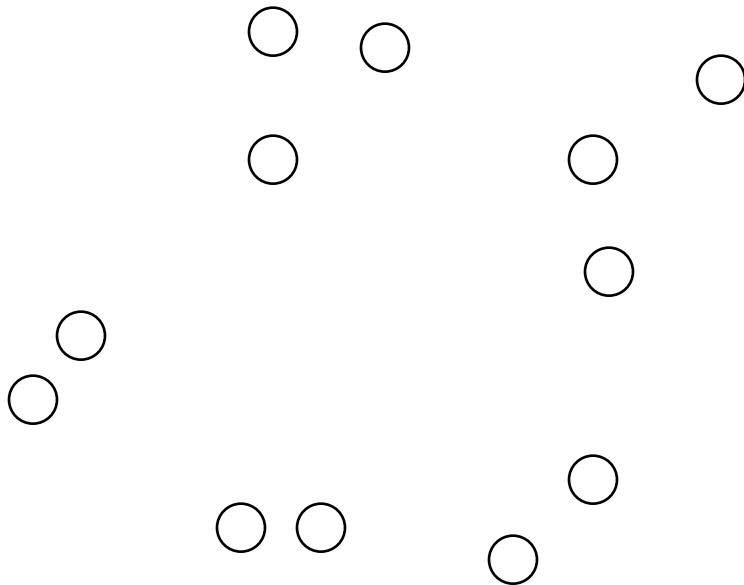


Agglomerative Clustering Algorithm

- More popular hierarchical clustering technique
- Basic algorithm is straightforward
 1. Compute the distance matrix
 2. Let each data point be a cluster
 3. **Repeat**
 4. Merge the two closest clusters
 5. Update the distance matrix
 6. **Until** only a single cluster remains
- Key operation is the computation of the distance between two clusters
 - Different approaches to defining the distance between clusters distinguish the different algorithms

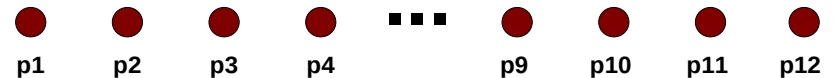
Starting Situation

- Start with clusters of individual points and a distance matrix



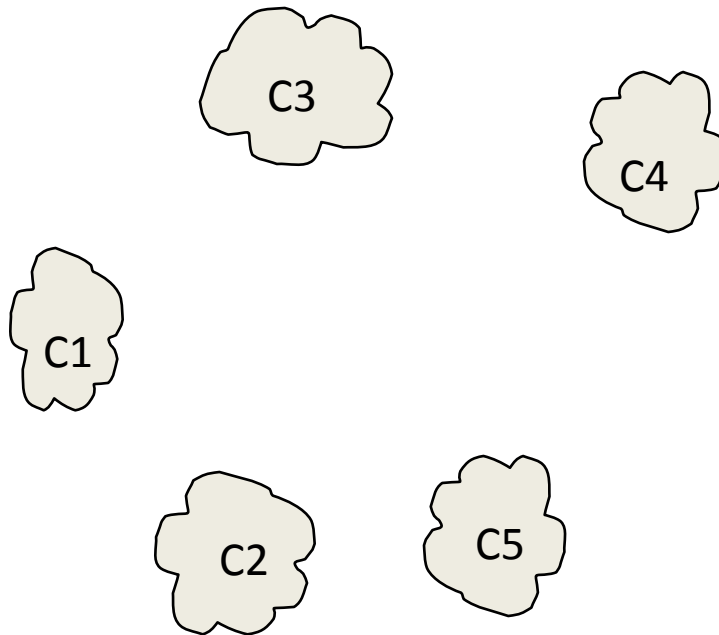
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Distance Matrix



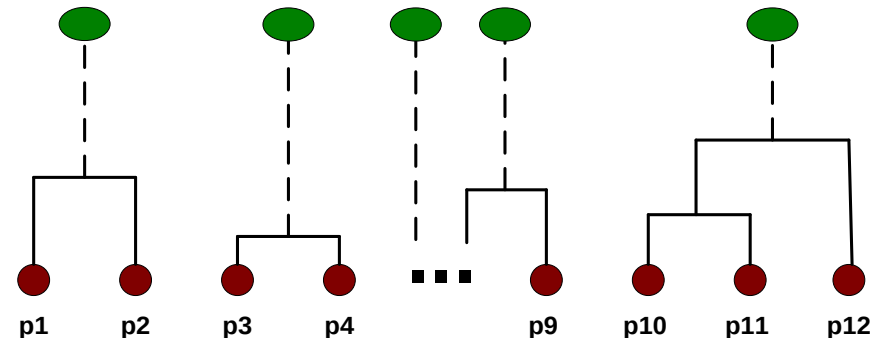
Intermediate Situation

- After some merging steps, we have some clusters
- Choose two clusters that has the smallest distance (largest similarity) to merge



	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Distance Matrix

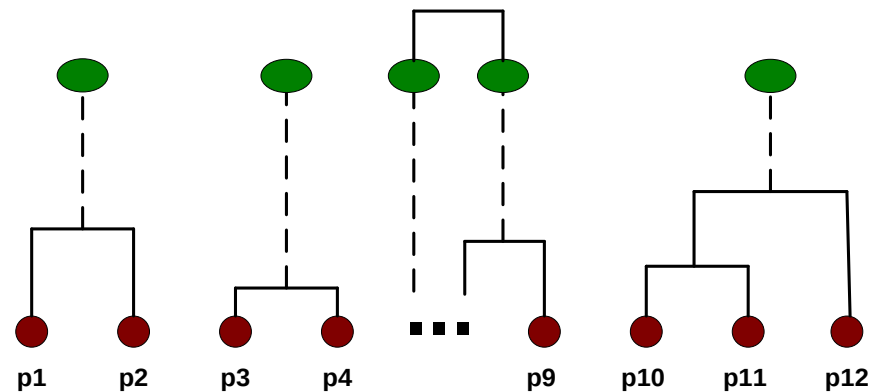
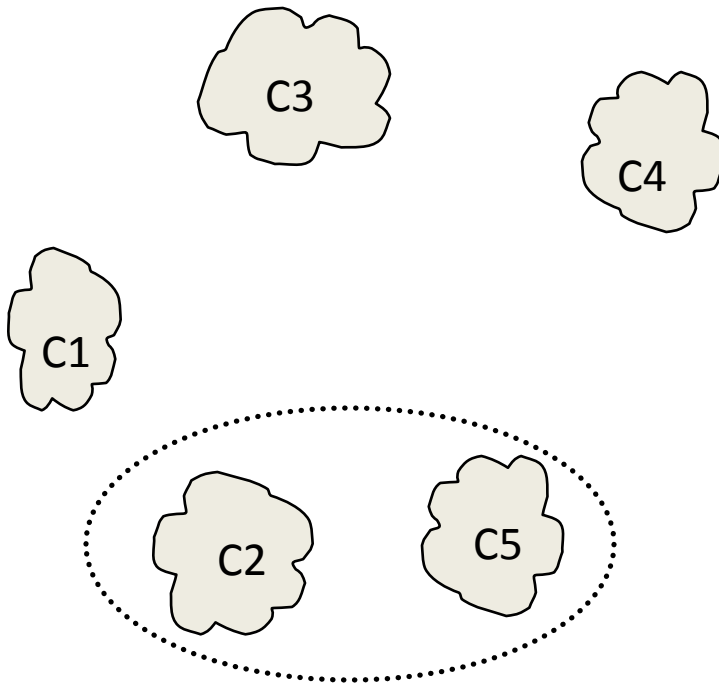


Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the distance matrix.

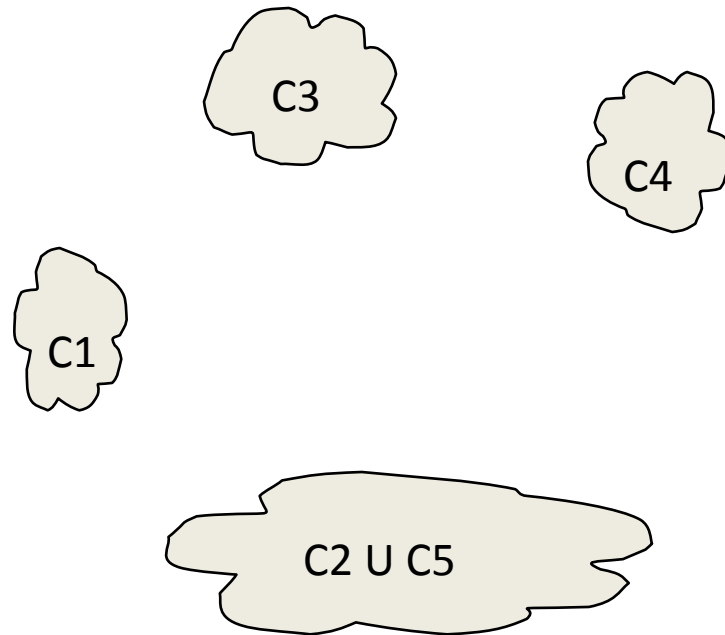
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Distance Matrix



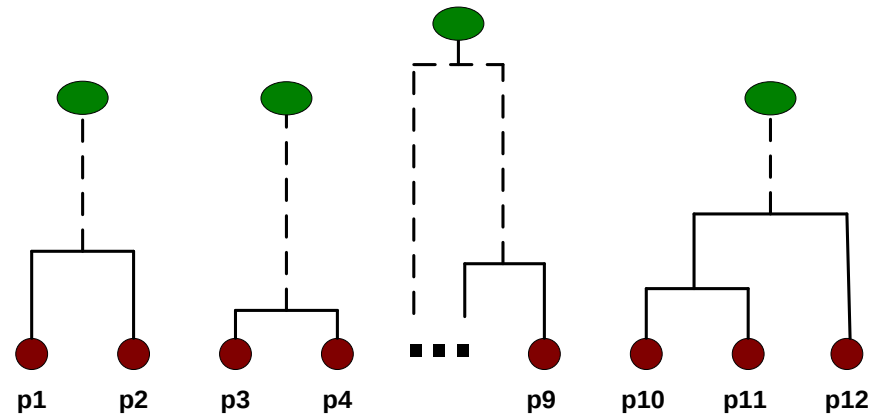
After Merging

- The question is “How do we update the distance matrix?”

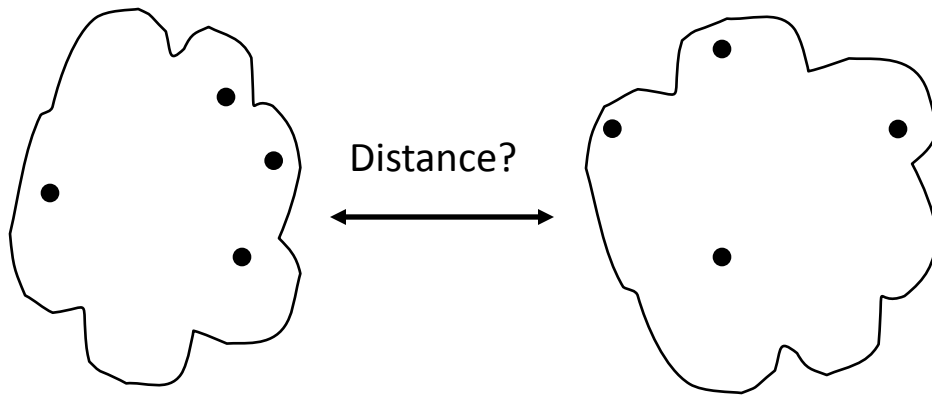


		C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Distance Matrix



How to Define Inter-Cluster Distance



- MIN
- MAX
- Group Average
- Distance Between Centroids
-

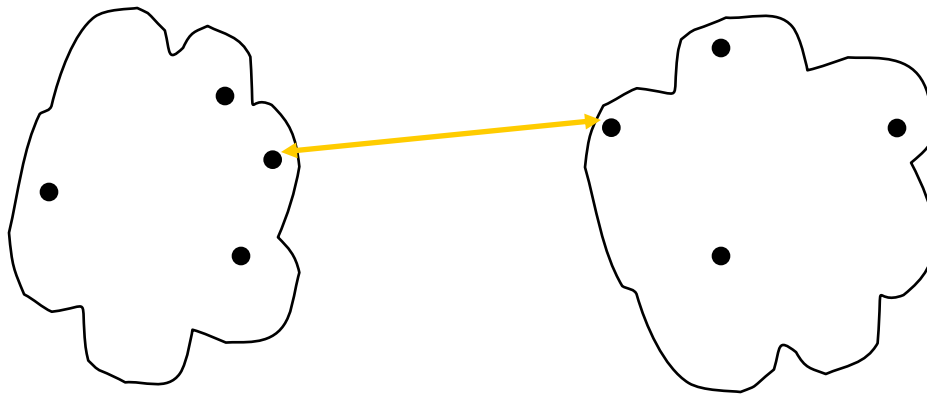
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

· Distance Matrix
·

MIN or Single Link

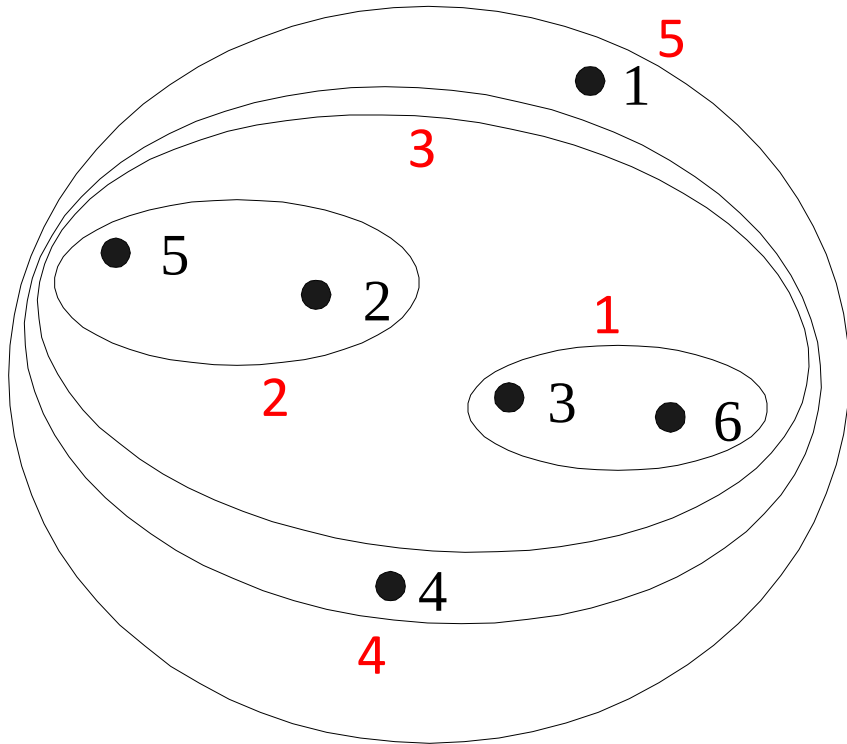
- **Inter-cluster distance**

- The distance between two clusters is represented by the distance of the closest pair of data objects belonging to different clusters.
- Determined by one pair of points, i.e., by one link in the proximity graph

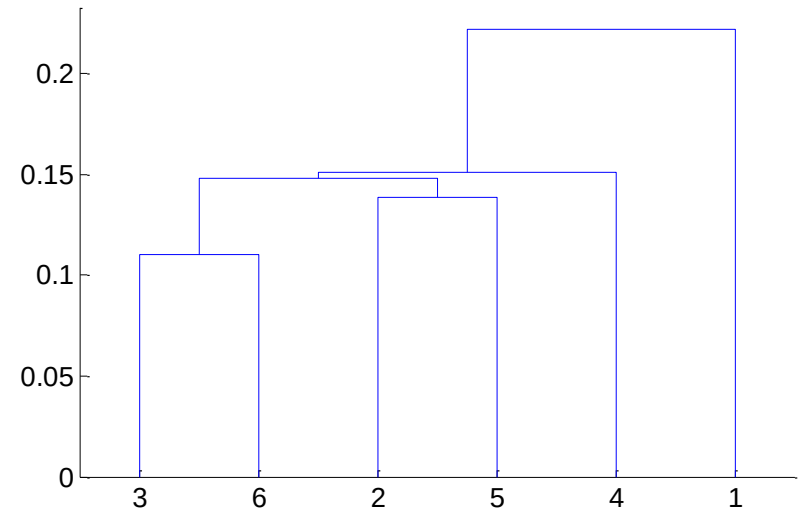


$$d_{\min}(C_i, C_j) = \min_{p \in C_i, q \in C_j} d(p, q)$$

MIN

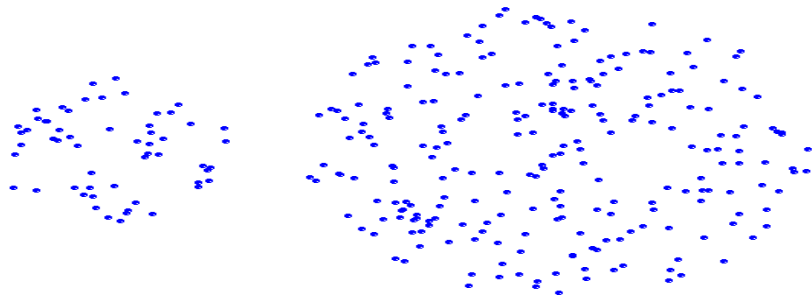


Nested Clusters

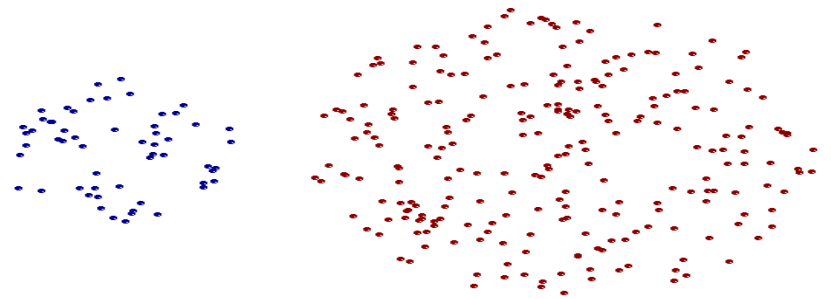


Dendrogram

Strength of MIN



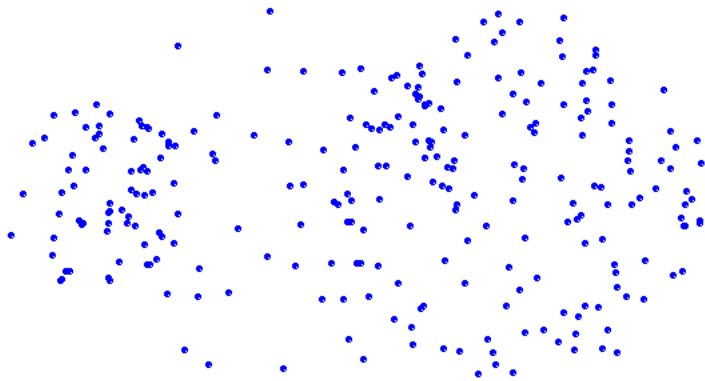
Original Points



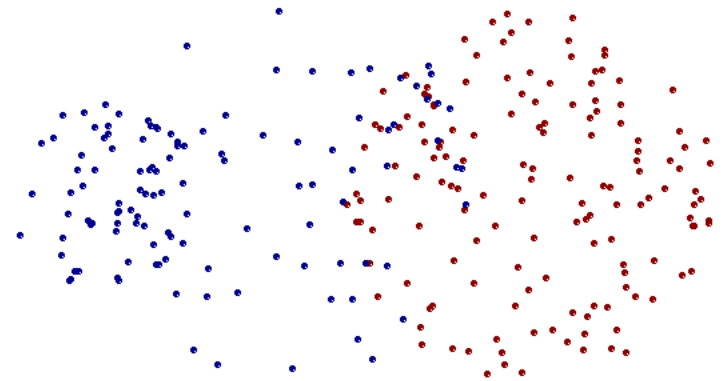
Two Clusters

- Can handle non-elliptical shapes

Limitations of MIN



Original Points



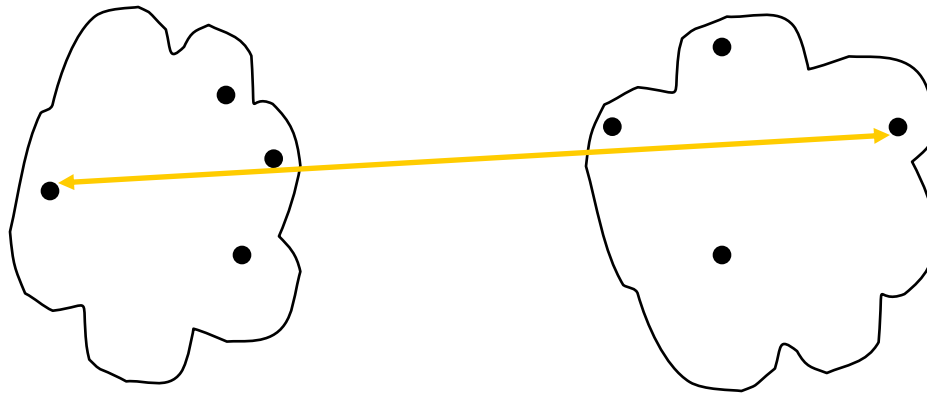
Two Clusters

- Sensitive to noise and outliers

MAX or Complete Link

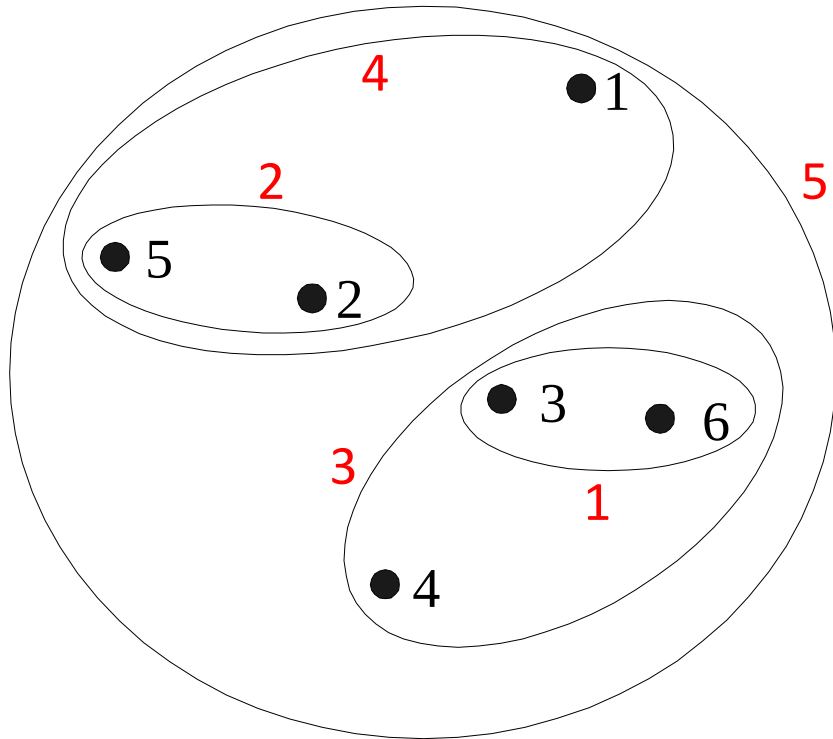
- **Inter-cluster distance**

- The distance between two clusters is represented by the distance of the farthest pair of data objects belonging to different clusters

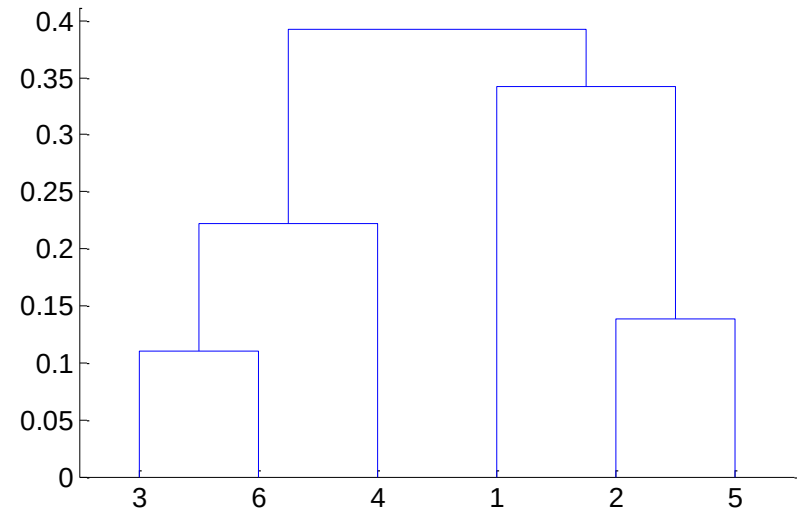


$$d_{\min}(C_i, C_j) = \max_{p \in C_i, q \in C_j} d(p, q)$$

MAX

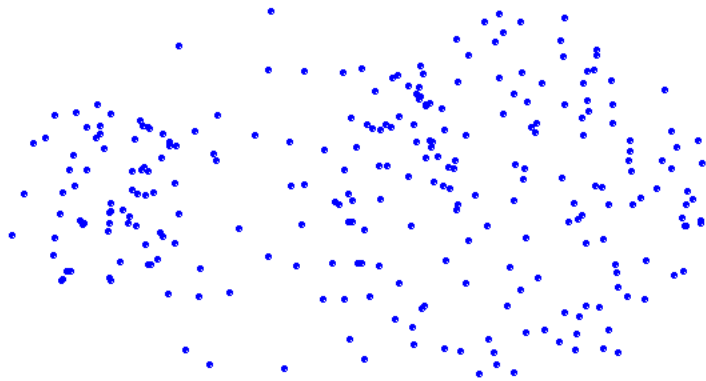


Nested Clusters

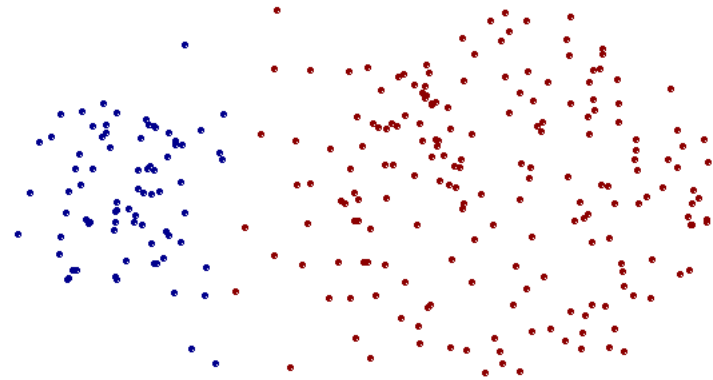


Dendrogram

Strength of MAX



Original Points

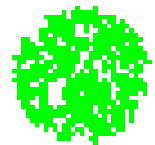
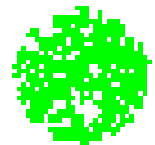
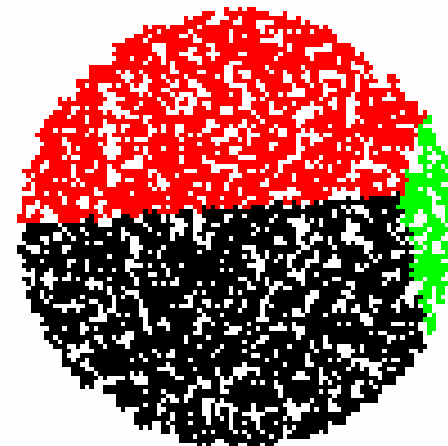
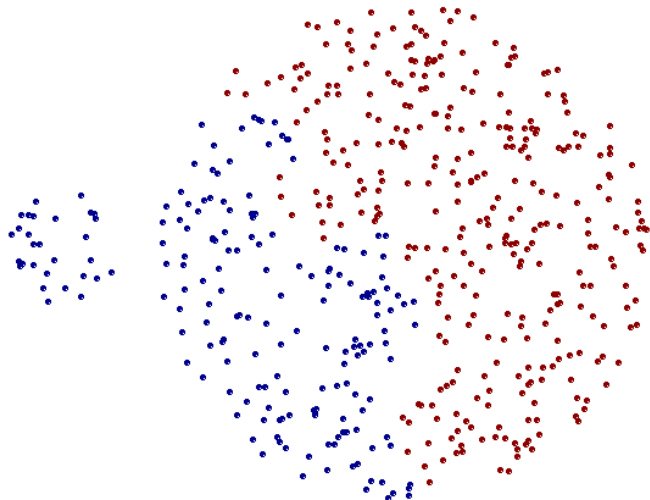
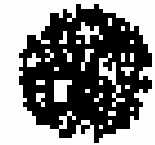
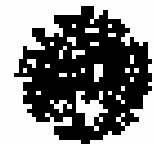
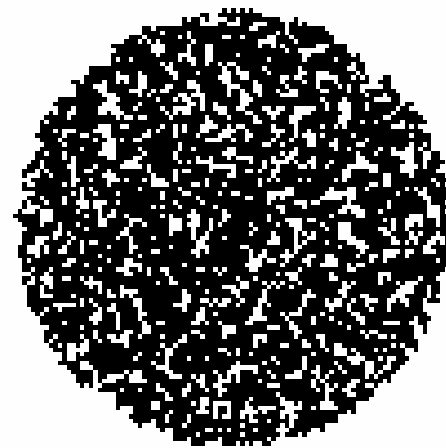
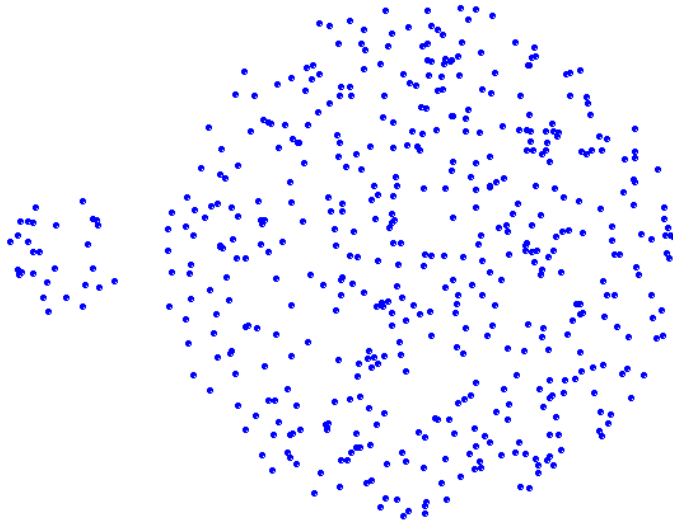


Two Clusters

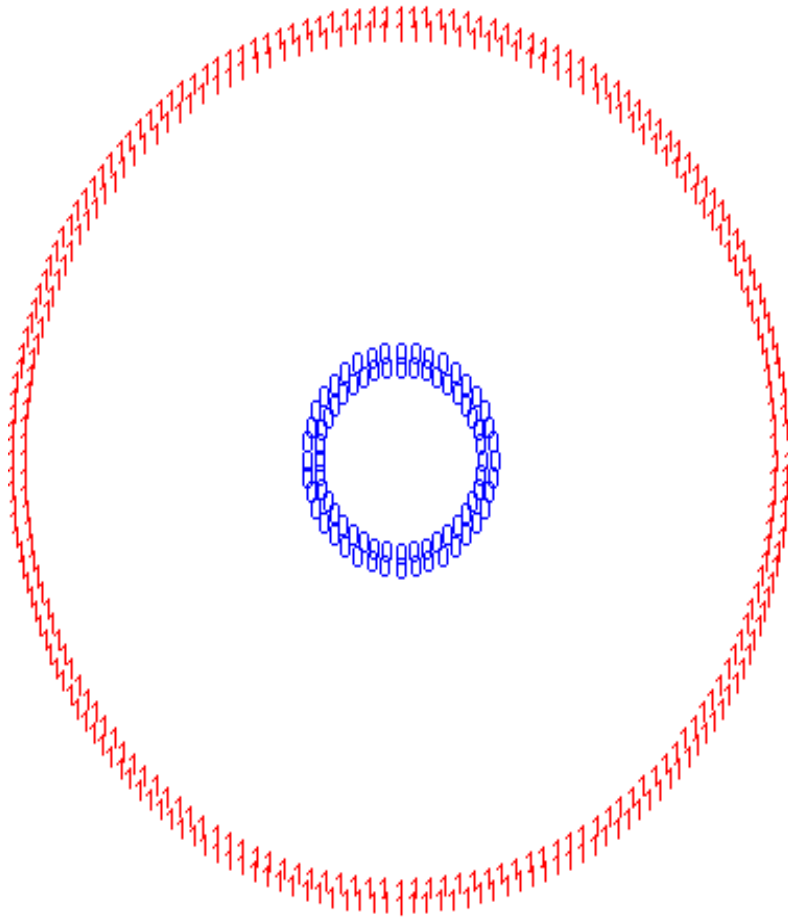
- Less susceptible to noise and outliers

Limitations of MAX

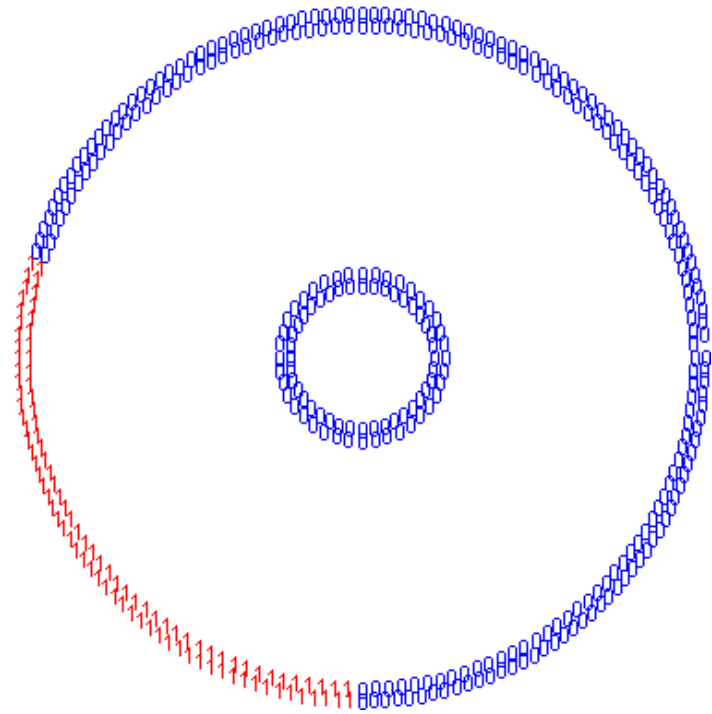
- Tends to break large clusters



Limitations of MAX



MIN (2 clusters)



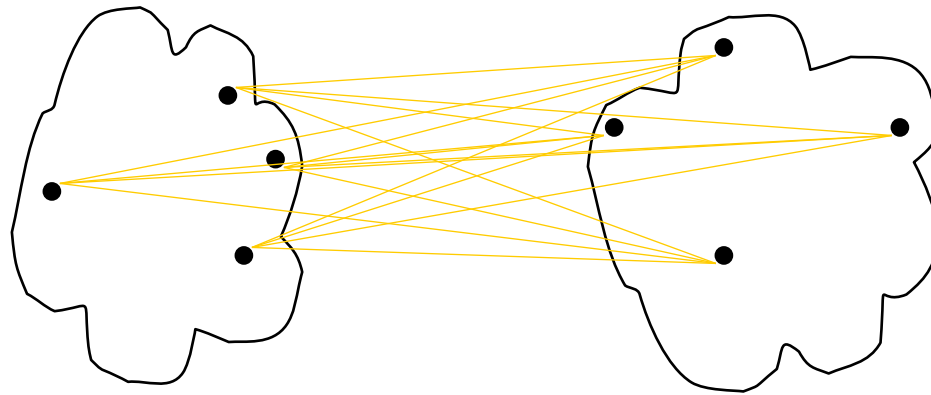
MAX (2 clusters)

- Biased towards globular clusters

Group Average or Average Link

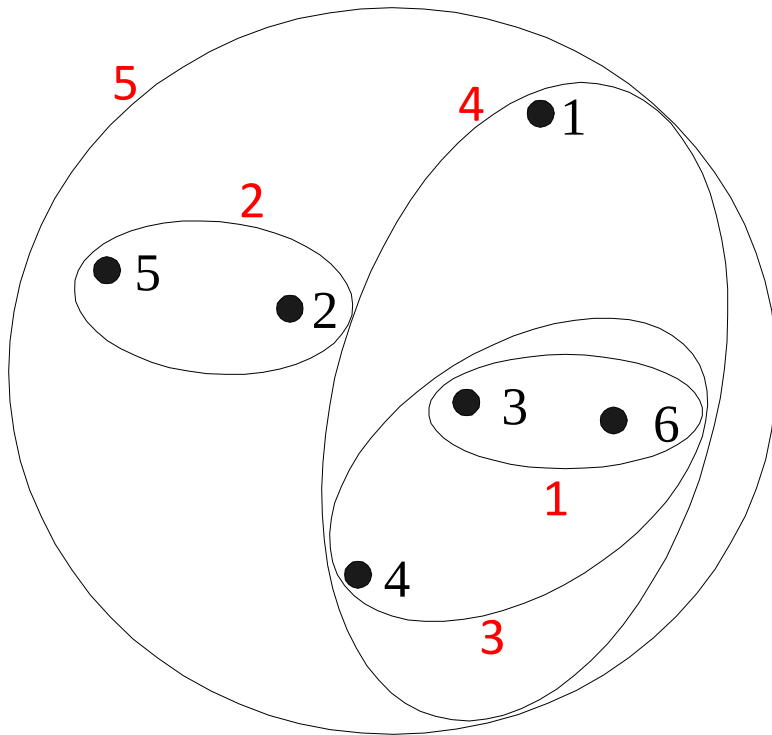
- **Inter-cluster distance**

- The distance between two clusters is represented by the average distance of all pairs of data objects belonging to different clusters
- Determined by all pairs of points in the two clusters

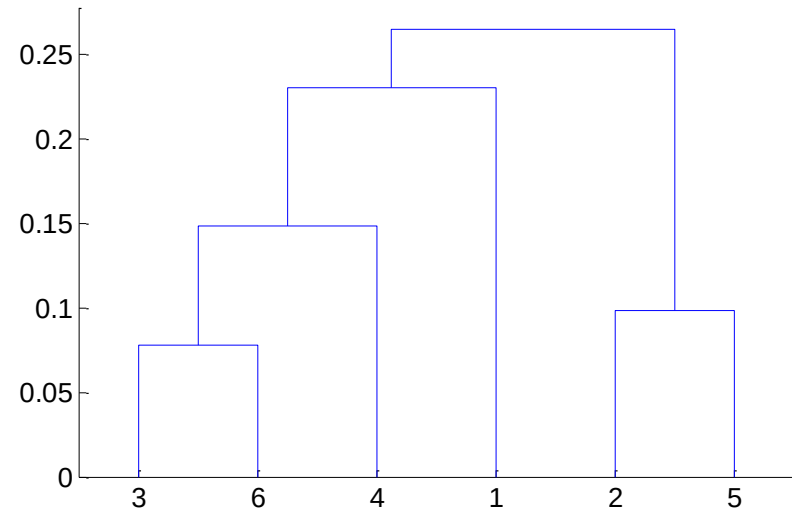


$$d_{\min}(C_i, C_j) = \text{avg}_{p \in C_i, q \in C_j} d(p, q)$$

Group Average



Nested Clusters



Dendrogram

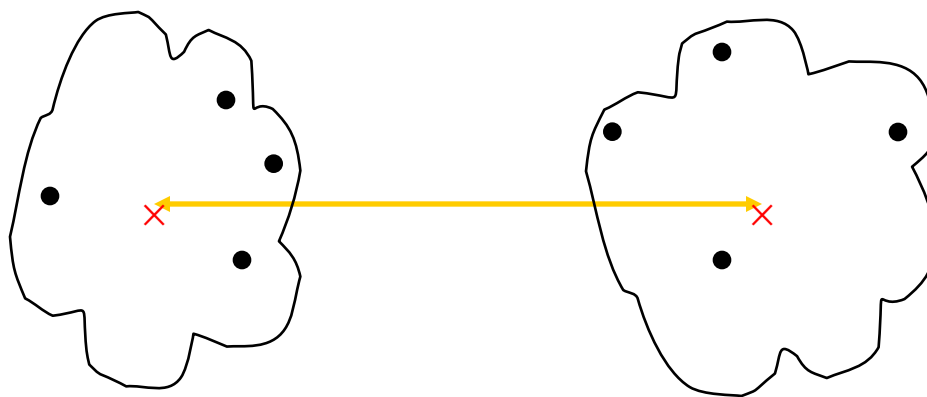
Group Average

- Compromise between Single and Complete Link
- Strengths
 - Less susceptible to noise and outliers
- Limitations
 - Biased towards globular clusters

Centroid Distance

- **Inter-cluster distance**

- The distance between two clusters is represented by the distance between the centers of the clusters
- Determined by cluster centroids

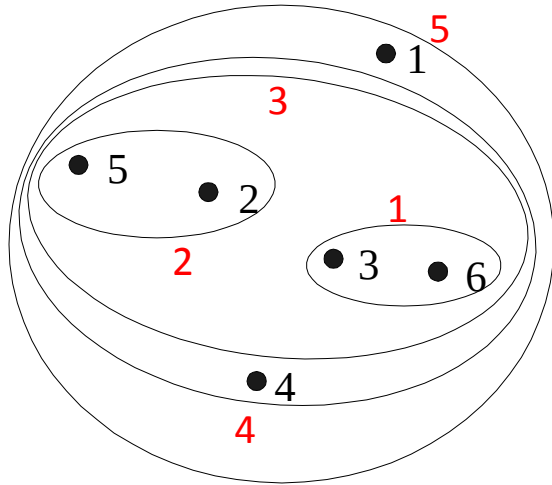


$$d_{mean}(C_i, C_j) = d(m_i, m_j)$$

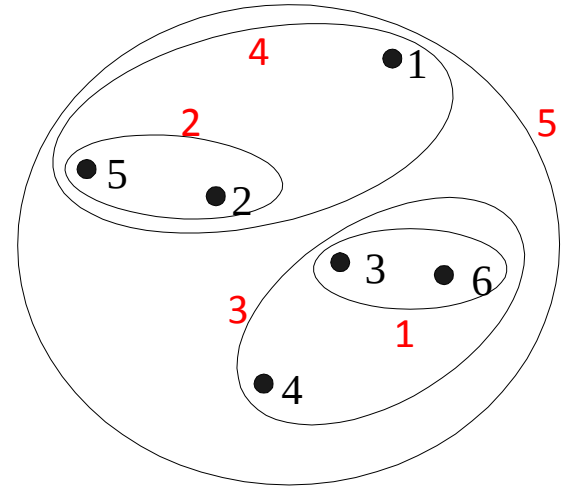
Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
 - Similar to group average if distance between points is distance squared
- Less susceptible to noise and outliers
- Biased towards globular clusters
- Hierarchical analogue of K-means
 - Can be used to initialize K-means

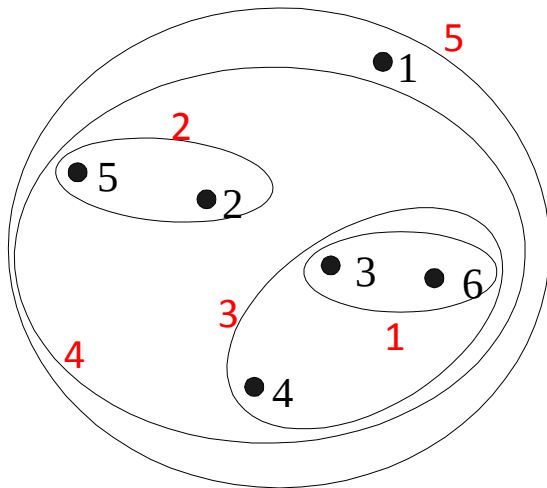
Comparison



MIN

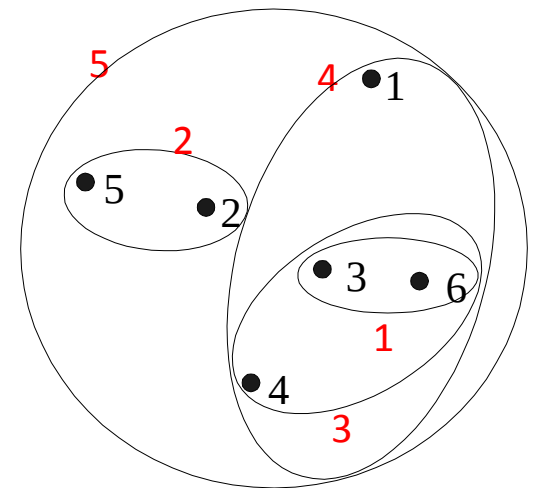


MAX



Group Average

Ward's Method



Time and Space Requirements

- $O(N^2)$ space since it uses the distance matrix
 - N is the number of points
- $O(N^3)$ time in many cases
 - There are N steps and at each step the size, N^2 , distance matrix must be updated and searched
 - Complexity can be reduced to $O(N^2 \log(N))$ time for some approaches

Strengths

- Do not have to assume any particular number of clusters
 - Any desired number of clusters can be obtained by ‘cutting’ the dendrogram at the proper level
- They may correspond to meaningful taxonomies
 - e.g., shopping websites—electronics (computer, camera, ..), furniture, groceries

Problems and Limitations

- Once a decision is made to combine two clusters, it cannot be undone
- No objective function is directly minimized
- Different schemes have problems with one or more of the following:
 - Sensitivity to noise and outliers
 - Difficulty handling different sized clusters and irregular shapes
 - Breaking large clusters

Take-away Message

- Agglomerative and divisive hierarchical clustering
- Several ways of defining inter-cluster distance
- The properties of clusters outputted by different approaches based on different inter-cluster distance definition
- Pros and cons of hierarchical clustering